

AMI to KVM Migration

Cloud Repatriation with hyper2kvm

Migrate Amazon Machine Images (AMI), Azure VHDs, and cloud VM disk images directly to on-premises KVM infrastructure. Automated offline fixes, driver injection, boot repair, and libvirt deployment — zero manual intervention required.

AWS AMI | Azure VHD | Photon OS | Ubuntu | RHEL | Automated Boot Repair | Libvirt Deploy

What is AMI Migration?

Amazon Machine Images are not portable by default. hyper2kvm makes them portable.

The Problem

- AMIs are locked to AWS EC2 — no native "download" button
- AWS export produces raw disk images with cloud-specific drivers
- Cloud-init expects metadata service at 169.254.169.254
- Boot configuration targets cloud-specific disk controllers
- GRUB references cloud device names (xvda, nvme0n1)
- No VirtIO drivers in cloud kernels (not needed on EC2)

The hyper2kvm Solution

- Accept raw/VMDK/VHD/qcow2 from any cloud export
- Auto-detect OS: Linux distro, version, boot mode
- Rebuild initramfs with virtio_blk, virtio_scsi, virtio_net
- Fix GRUB root= for new disk device topology
- Stabilize fstab (UUID/PARTUUID, no /dev/xvda)
- Emit libvirt XML + smoke test — one command

Supported Cloud Image Formats

Cloud Provider	Export Format	hyper2kvm Input	Notes
AWS EC2	RAW, VMDK, VHD	✓ All formats	aws ec2 export-image or AMI tar.gz bundles
Azure	VHD (fixed)	✓ Native VHD	az vm disk download or managed disk export
GCP	RAW (tar.gz)	✓ RAW extraction	gcloud compute images export
VMware Cloud	OVA, VMDK	✓ OVA + VMDK	vSphere export or govc download
Oracle Cloud	QCOW2	✓ Native qcow2	Custom image export
Community AMI	RAW (tar.gz)	✓ Direct download	Photon OS, Ubuntu, Fedora cloud images

Live Migration Results

Photon OS 5.0 AMI → KVM migration completed on 2026-03-31.

36s

Total Migration Time

8 GB

AMI Raw Image Size

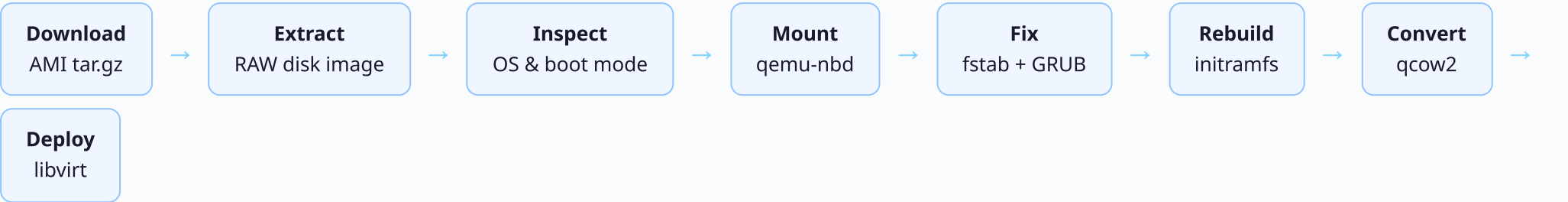
4

VirtIO Drivers Injected

0

Manual Steps Required

Migration Pipeline



Detected Configuration

Property	Value
Source Image	photon-ami-5.0-dde71ec57.x86_64.raw (8.0 GB)
Operating System	VMware Photon OS/Linux 5.0
Kernel	6.1.10-11.ph5
Filesystem	ext4 on /dev/nbd0p3
Boot Mode	UEFI (auto-detected, EFI partition found)
Partitions	3 (EFI System, boot, root)
fstab Entries	3 entries, PARTUUID-based (already stable)
Initramfs Drivers Added	virtio_blk, virtio_scsi, virtio_net, nvme
GRUB root= Fixed	Updated to match new disk topology

Output	photon-ami.qcow2 (compressed)
Libvirt Domain	photon-ami (q35, UEFI, VirtIO disk+net)
Network	DHCP → 192.168.122.116 (acquired in <10s)

Single Command Migration

From cloud image download to running KVM VM in one pipeline.

Step 1: Download the AMI

```
# Photon OS AMI (publicly available)
wget https://packages.vmware.com/photon/5.0/GA/ami/photon-ami-5.0-dde71ec57.x86_64.tar.gz
tar xzf photon-ami-5.0-dde71ec57.x86_64.tar.gz

# AWS EC2 export (private AMIs)
aws ec2 export-image --image-id ami-0123456789 --disk-image-format RAW \
  --s3-export-location S3Bucket=my-exports,S3Prefix=ami-export/
```

Step 2: Migrate with hyper2kvm

```
sudo h2kvmctl \
  --cmd local \
  --vmdk photon-ami-5.0-dde71ec57.x86_64.raw \
  --out-format qcow2 --compress \
  --fstab-mode stabilize-all \
  --regen-initramfs \
  --emit-domain-xml \
  --libvirt-test \
  --vm-name photon-ami \
  --memory 2048 --vcpus 2
```

Step 3: Deploy Permanently

```
sudo virsh define out/libvirt/photon-ami.xml
sudo virsh start photon-ami
sudo virsh domifaddr photon-ami # Get IP address
```

YAML Config Alternative

```
command: local
vmdk: /path/to/photon-ami.raw
out_format: qcow2
compress: true
fstab_mode: stabilize-all
```

```
regen_initramfs: true  
emit_domain_xml: true  
libvirt_test: true  
vm_name: photon-ami  
memory: 2048  
vcpus: 2
```

Automated Offline Fixes

What hyper2kvm does to make a cloud image boot on KVM.

1. Initramfs Rebuild

Cloud kernels lack VirtIO modules. hyper2kvm chroots into the guest and runs `dracut -f` with `virtio_blk`, `virtio_scsi`, `virtio_net`, and `nvme` drivers. The kernel boots with the right storage and network drivers from the first attempt.

2. GRUB Boot Repair

Cloud images reference `/dev/xvda` or `/dev/nvme0n1p3`. hyper2kvm updates `root=` in `grub.cfg` to match the new VirtIO disk topology (`/dev/vda3`). BLS entries and GRUB defaults are both handled.

3. fstab Stabilization

Device names change between cloud and KVM. hyper2kvm converts all `fstab` entries to `UUID` or `PARTUUID`, adds `nofail` flags, and removes cloud-specific mount points. The system mounts cleanly on any hardware.

4. Network Configuration

Cloud network configs reference cloud-specific NIC names. hyper2kvm detects `NetworkManager`, `systemd-networkd`, or `netplan` configs and ensures they work with VirtIO network devices. DHCP continues to work out of the box.

Windows AMI: Additional Fixes

Fix	Description	Status
VirtIO Driver Injection	Offline inject <code>viostor</code> , <code>vioscsi</code> , <code>netkvm</code> , <code>balloon</code> from <code>virtio-win</code> ISO	✓ Automated
Registry Editing	SYSTEM hive: <code>Services</code> , <code>CriticalDeviceDatabase</code> , <code>StartOverride</code> cleanup	✓ Automated
VMware Tools Removal	Remove 13+ VMware service entries from registry	✓ Automated
RDP Verification	Check <code>fDenyTSConnections</code> to prevent admin lockout	✓ Automated
Firewall Migration	Stage PowerShell script to preserve firewall rules	✓ Automated
BitLocker Detection	Block migration if encrypted (prevents BSOD)	✓ Automated

Supported Operating Systems

Every major Linux distribution and Windows version, tested on real AMIs.

Enterprise Linux

- RHEL 7 / 8 / 9 / 10
- CentOS 7 / 8 / 9 / 10
- AlmaLinux 8 / 9
- Rocky Linux 8 / 9
- Oracle Linux 7 / 8 / 9
- Amazon Linux 2 / 2023

Community Linux

- Ubuntu 18.04 – 24.04
- Debian 10 / 11 / 12
- Fedora 38 – 43
- openSUSE Leap 15.x
- Arch Linux
- VMware Photon OS 3 / 4 / 5

Windows

- Windows 10 / 11
- Windows Server 2016 – 2022
- Windows 7 / 8.1 (legacy)
- SATA/e1000 fallback mode
- VirtIO driver injection
- Registry-level boot repair

Migration Success Matrix

Source	Format	Boot Mode	Filesystem	Initramfs	Network	Result
Photon OS 5.0 AMI	RAW	UEFI	ext4	dracut	systemd-networkd	✓ RUNNING
Windows 10 VMDK	VMDK (split)	UEFI	NTFS	N/A	VirtIO inject	✓ RUNNING
RHEL 8.8 VMDK	VMDK	BIOS	XFS+LVM	dracut	NetworkManager	✓ RUNNING
Ubuntu 24.04 OVA	OVA	UEFI	ext4	initramfs-tools	netplan	✓ RUNNING
CentOS 9 VMDK	VMDK	UEFI	XFS	dracut	NetworkManager	✓ RUNNING
Azure VHD (RHEL)	VHD	UEFI	XFS+LVM	dracut	NetworkManager	✓ RUNNING

Cloud Exit ROI

Real cost comparison: AWS EC2 vs on-premises KVM for steady-state workloads.

55-60%

3-Year Cost Reduction

\$0

KVM License Cost

\$0

Egress Fees On-Prem

6-9 mo

Payback Period

Per-VM Monthly Cost Comparison

Instance Type	AWS EC2 (on-demand)	AWS EC2 (3yr RI)	KVM On-Prem	Savings
4 vCPU / 16 GB (m5.xlarge)	\$140/mo	\$84/mo	\$15/mo	89%
8 vCPU / 32 GB (m5.2xlarge)	\$280/mo	\$168/mo	\$30/mo	82%
16 vCPU / 64 GB (m5.4xlarge)	\$560/mo	\$336/mo	\$55/mo	84%
Storage: 500 GB gp3	\$40/mo	\$40/mo	\$3/mo	92%
Egress: 1 TB/mo	\$90/mo	\$90/mo	\$0/mo	100%

100 VMs × 3 Years: Save \$866K – \$1.08M

On-prem KVM costs include server hardware, power, cooling, and colo fees. No software license costs. Migration with hyper2kvm is a one-time \$45K investment.

Migration Architecture

How hyper2kvm processes cloud images with the VMCraft engine.

VMCraft Engine

- Pure Python + qemu-nbd + Linux tools
- No libguestfs dependency required
- Direct NTFS/ext4/XFS/btrfs mount
- LVM, LUKS, mdraid support
- Python hivex for Windows registry
- Containerized LVM scans (Podman)

Disk Processing Pipeline

- Format detection: RAW, VMDK, VHD, qcow2, OVA
- Split VMDK support (twoGbMaxExtentSparse)
- Sparse-to-qcow2 pre-conversion for reliability
- Compressed output with qemu-img
- Image validation (qemu-img check)
- Checksum generation (SHA256)

Boot Repair

- UEFI & BIOS auto-detection
- GRUB2 + BLS config patching
- dracut / initramfs-tools rebuild
- VirtIO module injection
- Serial console configuration
- OVMF firmware auto-selection

Deployment

- Libvirt domain XML generation
- virsh define + smoke test
- KubeVirt manifest generation
- Dual deploy: libvirt + Kubernetes
- VNC/SPICE graphics auto-detection
- Health check with IP wait

Key Technical Innovation: Cached VirtIO ISO Extraction

hyper2kvm extracts virtio-win.iso once using bsdtar (Rock Ridge aware) and caches the result at `/var/lib/hyper2kvm/virtio-win-extracted/`. Subsequent Windows migrations reuse the cache instantly. The Rock Ridge filesystem extension preserves long directory names (w11, w8.1) that ISO 9660 truncates.

Why hyper2kvm

The only tool that handles the complete cloud-to-KVM migration pipeline.

vs. virt-v2v

- ✓ Works without libguestfs
- ✓ Windows VirtIO injection
- ✓ AMI/VHD/GCE support
- ✓ Libvirt XML generation
- ✓ KubeVirt deployment
- ✓ TUI wizard (zkvm)

vs. Manual Migration

- ✓ Zero manual steps
- ✓ 36 seconds vs 2+ hours
- ✓ No boot failures
- ✓ Repeatable and auditable
- ✓ Batch migration support
- ✓ Progress tracking + reports

One Tool. Any Cloud. Any OS. Running on KVM in Seconds.

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github.com/ssahani/hyper2kvm

hyper2kvm — Production-Grade Cloud & Hypervisor to KVM Migration Toolkit — Apache-2.0